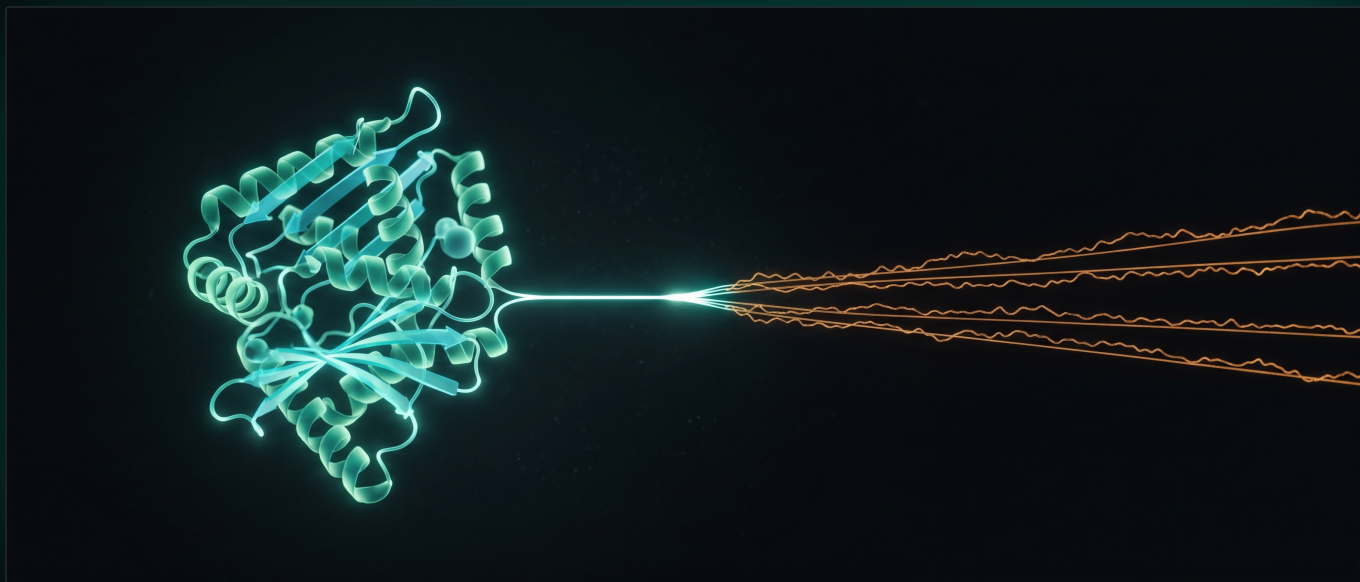




Judge Prep

Pravin Ireri

ENGINE A CLEAVABLE FIBER • ENGINE B MATCHED ENZYME • THE MOLECULAR
SCIENCE

You own the science the grand-prize judge will probe hardest. Every fiber and enzyme value here is verified against the artifacts. The single rule that protects the whole pitch: the enzymes are honest LigandMPNN redesigns, not de-novo folds, and you say that proudly. An invented de-novo claim is the one thing that loses this room.

— 01 THE THIRTY-SECOND VERSION

REWEAR-FUSED is **one computational co-design loop** that designs two matched molecules at once to close the textile-to-textile recycling loop on children's stretch apparel, non-recyclable today because of its elastane.

Instead of attacking existing elastane (crystalline, hostile, it defeats every native enzyme on the intact fiber), we invert the problem. **Engine A**, your lane, screens a new, enzymatically-**cleavable** elastane fiber. **Engine B**, your lane, designs the **matched** carbamate hydrolase built to cut that exact bond. **Engine C** is the live Digital Recyclability Passport.

"Carter's, here's the molecule for your next onesie, and the matched enzyme that closes its loop. No one has co-designed the pair, together, for textiles."

```
GARMENT → ENGINE C passport: clear / lab-test / divert
      ↳ ENGINE A cleavable elastane (constraint-guided screening) ← you
            ↳ [carbamate substrate target] → ENGINE B matched hydrolase ← you
                  ↳ cleaves bond → recoverable monomers → NEW onesie
```

— 02 THE TWO WALLS (WHY THIS EXISTS)

- **Wall 1, chemical safety.** Nobody can prove recovered fabric is safe for infant skin. (Engine C.)
- **Wall 2, elastane.** Nobody can break down the elastane in every stretch garment. **This is the wall your two engines knock down.**

LEAD STAT

Elastane is in **~80%** of clothing, and **~1%** of it makes a recycler reject the **whole** garment. (Fashion for Good, 2026.)

SCALE

~92M tonnes textile waste a year (UNEP); only **~1%** recycled fiber-to-fiber.

03 WHAT IS REAL, WHAT IS IN-SILICO (THE HONESTY LEDGER)

COMPONENT	TIER	TRL	WHAT IT IS
Engine C, the Passport	WIRED LIVE	6	real FastAPI endpoint + deployed UI
The app (5 views)	WIRED LIVE	-	on Vercel, reading the signed bundle
Engine A, the fiber	IN-SILICO	3	constraint-guided screening; predicted properties, benchmark-referenced
Engine B, the enzyme	IN-SILICO	2	LigandMPNN redesign (TM ~0.8); de-novo fold is the deferred swing
Wet-lab validation	FORWARD	-	the Cox Cleantech Residency ask

04 THE DE-NOVO QUESTION (YOUR #1 GUARDRAIL, MEMORIZE IT)

IF YOU SAY ONE THING RIGHT TODAY, SAY THIS

"Honestly, no, we did not design a de-novo enzyme yet, and we do not claim to. What we ship and render are three **LigandMPNN redesigns** of real serine-hydrolase scaffolds (IsPETase, FAST-PETase, CalB), in-silico, FoldSeek TM ~0.8 to template. The true de-novo fold, no prior art, TM < 0.5, is the GPU swing we scoped and deferred. We even built a swap script that **refuses any structure with TM ≥ 0.5**, so we cannot overclaim. The 'shouldn't be possible' novelty is the **co-design loop**: the matched fiber-and-enzyme pair, which is verified open. A faked de-novo would have been our loss condition."

This is more credible than a de-novo claim a judge could puncture. The redesign is a real artifact. The honesty is the moat.

— 05 THE FIVE SCIENTIFIC CONSTRAINTS (YOU ARE THE SCIENCE ANSWER)

#	CONSTRAINT	THE WHY (YOUR DOMAIN)
1	Aliphatic isocyanates only	aromatic ones cleave to MDA, IARC 2B carcinogen, REACH Annex XVII, OEKO-TEX Class I. Aliphatic gives benign aliphatic diamines.
2	Ser-His-Asp, pH 7.0-8.5	wild-type UMG-SP2 (Ser-Ser-cis-Lys) needs pH ~10, which damages cotton. We anchor on its transition-state geometry, not its protein.
3	Cleavable bond in soft segment; 25-35 wt% hard	crystalline hard domains carry the load; cleavable bond in the amorphous phase keeps >400% stretch. <17 too fast, >40 won't degrade.
4	Screening, not generative	polymer LMs (polyBART/polyBERT) only handle homopolymers + random copolymers, not segmented block PU-ureas. Enumerate, score, filter.
5	In-silico, benchmark-referenced	nothing spun or expressed. Cyclic fatigue / elastic recovery / hysteresis are not ML-predictable and need wet lab. Say so.

— 06 DO NOT SAY (THE OVER-CLAIMS THAT LOSE THE ROOM)

- ✗ **Not "de-novo enzyme" / "novel fold" for the shipped designs.** LigandMPNN redesigns, TM 0.79-0.85. De-novo is deferred. Claim de-novo only for the co-design loop.
- ✗ **Not "the structure shown is the designed protein".** It is the real CalB / PDB 1TCA reference scaffold the lead redesign is built on, shown as a reference; the designed .bcif is a download.
- ✗ **Not measured PLACER outputs.** They are in-silico design targets on the Lauko thresholds. The UI says "Design metrics, not an expressed protein."
- ✗ **No kcat / KM / kcat/KM.** None exists; urethane hydrolysis is harder than the ester chemistry those figures come from.
- ✗ **Not "generative design of a copolymer".** Constraint-guided screening.
- ✗ **Not "the wild-type triad is Ser-His-Asp".** Wild-type UMG-SP2 is Ser-Ser-cis-Lys at pH ~10; we designed the Ser-His-Asp near-neutral triad.
- ✗ **Not "72 hours" / build-duration (DQ risk).** No DOI outside the verified section-9 list.

07 ENGINE A · CLEAVABLE ELASTANE (CONSTRAINT-GUIDED SCREENING)

You enumerate candidate polyurethane-urea compositions from known building blocks, score them with property surrogates, and filter to the design window. Not generative. Screening.

THE FUNNEL (VERIFIED IN SCREENING.JSON)

612 enumerated → 487 scored → 23 filtered → 10 top candidates.

BUILDING BLOCK	WHAT WE USE (ALIPHATIC ONLY)
Isocyanates	HDI, IPDI, H12MDI
Soft diols (~2000 g/mol)	PCL-diol, Polycarbonate-diol
Chain extenders	BDO, EDA
Cleavage site (SMARTS)	[#6]-[NH]-C(=O)-O-[#6]

- All 10 candidates fall in the **25.8 to 33.4 wt%** hard-segment window (Constraint 3).
- Soft-segment Tg **-51 to -58 °C** (all < -50); elongation **440 to 620%** (all ≥ 400%).
- No aromatic isocyanate appears anywhere (Constraint 1; the validator rejects it).

THE TRADE-OFF CURVE (YOUR HONESTY CENTERPIECE)

Plots degradability vs mechanical retention as crossing curves across hard-segment wt% (10–50), with the **25–35** design window shaded and the annotation "<17 wt% degrades too fast; >40 wt% won't degrade". Cites Xu SusMat 2026 (10.1002/sus2.70038), Qin Adv. Sci. 2024 (10.1002/advs.202400255), Ding 2023 (10.1007/s10118-022-2838-6). It turns the mechanics-vs-cleavability tension into a demonstration of rigor.

— 08 ENGINE B • THE MATCHED CARBAMATE HYDROLASE

The pipeline spec: RFdiffusion2 → LigandMPNN → ESMFold → Boltz-2 → PLACER → FoldSeek. What shipped is the LigandMPNN-redesign stage; the full GPU run is the deferred swing.

DESIGN	SCAFFOLD (TEMPLATE)	FOLDSEEK TM	PLACER PREORG	PLDDT
enzyme-003 (featured)	CalB / PDB 1TCA	0.85	0.74	90.1
enzyme-001	IsPETase / 6EQE	0.82	0.71	88.2
enzyme-002	FAST-PETase / 7SH6	0.79	0.68	85.7

- All three: **Ser-His-Asp triad, pH 7.0-8.5**. enzyme-003 active-site pLDDT **95.2**, catalytic-Ca RMSD **0.38 Å**, scRMSD **1.18 Å**, all passing the Lauko thresholds (pLDDT>80, active-site>90, Ca-RMSD<1.0, scRMSD<2.0).
- **The .bcif files genuinely ARE their template scaffolds** (byte-verified: enzyme-003.bcif contains 1TCA/CANDIDA/LIPASE). The /enzyme view renders the real CalB/1TCA reference (residues 105/187/224 = Ser105-Asp187-His224); the designed enzyme-003.bcif is the download.
- **The anchor:** UMG-SP2 urethanase transition-state geometry (wild-type Ser-Ser-cis-Lys, most active pH ~10). We design a NEW near-neutral Ser-His-Asp triad. We never copy the wild-type protein.

THE HONESTY GATE, IN CODE

scripts/swap_denovo_enzyme.py is the one-command upgrade path for a real de-novo result. It **refuses any swap with TM ≥ 0.5** (only TM < 0.5 writes a "de-novo" verdict). The honesty gate is enforced in code, not just in copy. The manifest records model versions only for Engine C, which honestly reflects that the heavy GPU pipeline run is the deferred swing.

— 09 THE MATCHED PAIR (PROOF IT IS ONE LOOP)

Each enzyme's `matchedFiberId` points at a real fiber; `topPairs.json` carries the substrate SMILES of that fiber's carbamate microenvironment. The three matched pairs: `fiber-001 + enzyme-001`, `fiber-004 + enzyme-002`, `fiber-008 + enzyme-003` (that is the array order in `topPairs.json`; the first entry is `fiber-001 + enzyme-001`). The `/enzyme` view **features enzyme-003** (best PLACER score, the CalB/ITCA redesign), matched to `fiber-008`. "Featured" means best-scoring design, not "the first pair", keep those distinct if a judge asks. The matched pair is the defensible IP, composition-of-matter plus method, the white space no incumbent (Carbios, Samsara, Protein Evolution) occupies.

— 10 YOUR Q&A

Q Did you actually design a de-novo enzyme?

A Honestly, not yet, and we never claim it. We ship three LigandMPNN redesigns of real serine-hydrolase scaffolds (IsPETase 6EQE TM 0.82, FAST-PETase 7SH6 TM 0.79, CalB ITCA TM 0.85), in-silico. The de-novo fold (TM<0.5, no prior art) is the deferred GPU swing. We built the full pipeline and a swap script that refuses TM≥0.5 so we cannot overclaim. The redesign is real; an invented de-novo would have been the loss condition.

Q What structure am I looking at and downloading?

A The CalB / PDB ITCA serine-hydrolase reference scaffold, the real template enzyme-003 is a LigandMPNN redesign of. It is labeled as the reference, not an expressed protein. The file is the authentic ITCA crystal; the active-site highlight is its Ser105-Asp187-His224 triad.

Q Are your PLACER numbers measured?

A No, they are in-silico design targets on the Lauko (Science 2025) thresholds, reported with an uncertainty band. The lead shows preorg 0.74 +/- 0.03, pLDDT 90.1 (active-site 95.2), Ca-RMSD 0.38, scRMSD 1.18. The UI says "Design metrics, not an expressed protein." We ship no *koat*/KM because urethane hydrolysis is harder than the ester chemistry those numbers come from.

Q Can AI really invent a new block copolymer like elastane?

A We never claim that. Engine A is constraint-guided virtual screening: enumerate 612 from known aliphatic isocyanates (HDI/IPDI/H12MDI), soft diols (PCL/polycarbonate ~2000 g/mol), chain extenders (BDO/EDA), score 487, filter 23, surface 10. Polymer LMs only handle homopolymers and random copolymers, not segmented block PU-ureas.

Q Doesn't your enzyme need pH 10, which destroys the cotton?

A That's the wild-type UMG-SP2 problem (Ser-Ser-cis-Lys, pH ~10). We anchor on its transition-state geometry, not its protein, and design a Ser-His-Asp triad near-neutral. All three designs are Ser-His-Asp, pH 7.0-8.5; the validator rejects anything outside that band.

Q Won't making it cleavable destroy the stretch?

A No. The cleavable carbamate is in the soft amorphous segment, not the crystalline load-bearing domains. All 10 candidates sit in the 25-35 wt% window, elongation 440-620%, soft Tg below -50. The trade-off curve shows the window with "<17 too fast, >40 won't degrade", cited to Xu 2026 and Qin 2024.

Q What toxic products come off when you cleave it?

A None of concern. Aliphatic isocyanates only, which cleave to aliphatic diamines, not aromatic MDA/TDA. The validator rejects non-aliphatic isocyanates; the passport shows "aromatic amine release: NONE".

Q How do I know these numbers were not just typed in?

A Every artifact is SHA-256 signed in the manifest with producedBy provenance, and validate_artifacts.py --strict rejects a stale manifest on any byte change. We disclose the limit honestly: the manifest records a model version only for Engine C; there is no committed GPU-pipeline version for A/B because that heavy run is the deferred swing. We label the redesigns and design-target metrics as in-silico rather than dress them as committed outputs.

REWEAR-FUSED JUDGE PRAVIN · ENGINES A + B · HAND TO STEPHEN (FRONTEND) OR VINH (ENGINE C) OUTSIDE
PREP YOUR LANE